

1 The tensor product and the Kronecker product as one of its possible representations

1.1 The tensor product of vector spaces

Let $\underline{v} = \{v_i \mid i \in I\}$ and $\underline{w} = \{w_j \mid j \in J\}$ be bases for V and W . Then $V \otimes W$ is the product space, where there exists a basis which can be identified uniquely with the pairs of the cartesian product

$$\underline{v} \times \underline{w} = \{(v_i, w_j) \mid i \in I, j \in J\}, \quad (1)$$

so that

$$\dim(V \otimes W) = \dim V \cdot \dim W. \quad (2)$$

Since $V \otimes W$ is a vector space, an arbitrary element of it can be written as

$$\sum_{(i,j) \in I \times J} c_{ij} (v_i \otimes w_j). \quad (3)$$

The bilinear map $\otimes$ is defined by

$$\otimes: V \times W \longrightarrow V \otimes W, \quad (4)$$

$$(v, w) \longmapsto v \otimes w = \sum_{(i,j) \in I \times J} a_i b_j (v_i \otimes w_j), \quad (5)$$

for $v = \sum_i a_i v_i$ and $w = \sum_j b_j w_j$. As a check of bilinearity, writing $v' = \sum_i a'_i v_i$ and $v'' = \sum_i a''_i v_i$,

$$\begin{aligned} (v' + v'') \otimes w &= \left(\sum_i a'_i v_i + \sum_i a''_i v_i \right) \otimes w = \left(\sum_i (a'_i + a''_i) v_i \right) \otimes w \\ &= \sum_{i,j} (a'_i + a''_i) b_j (v_i \otimes w_j) \\ &= \sum_{i,j} a'_i b_j (v_i \otimes w_j) + \sum_{i,j} a''_i b_j (v_i \otimes w_j) \\ &= v' \otimes w + v'' \otimes w, \end{aligned} \quad (6)$$

and analogously in the second argument.

1.2 The tensor product of operators

Introduce operators $A: V \rightarrow V$ and $B: W \rightarrow W$ with V, W vector spaces. How is $A \otimes B: V \otimes W \rightarrow V \otimes W$ given? Define a map $\beta: V \times W \rightarrow V \otimes W$, $\beta(v, w) := Av \otimes Bw$. It is bilinear, e.g.

$$\begin{aligned} \beta(v_1 + v_2, w) &= A(v_1 + v_2) \otimes Bw = (Av_1 + Av_2) \otimes Bw \\ &\stackrel{\otimes \text{ bilin.}}{=} Av_1 \otimes Bw + Av_2 \otimes Bw = \beta(v_1, w) + \beta(v_2, w), \end{aligned} \quad (7)$$

and analogously in the second argument. By the universal property there exists a unique linear $\tilde{\beta}$ with $\tilde{\beta}(v \otimes w) := \beta(v, w)$, so that

$$(A \otimes B)(v \otimes w) = Av \otimes Bw. \quad (8)$$

1.3 The Kronecker product as a matrix representation

Let $V = K^m$ and $W = K^n$ and identify (this is *not* a unique representation!) $K^m \times K^n \rightarrow K^{(m \cdot n)}$ with the Kronecker product, i.e. for

$$v = \begin{pmatrix} a_1 \\ \vdots \\ a_m \end{pmatrix} \quad \& \quad w = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} \quad \text{set} \quad v \otimes w := \begin{pmatrix} a_1 w \\ \vdots \\ a_m w \end{pmatrix} = \begin{pmatrix} a_1 b_1 \\ \vdots \\ a_1 b_n \\ \vdots \\ a_m b_1 \\ \vdots \\ a_m b_n \end{pmatrix}. \quad (9)$$

Now choose $\{v_i\}$ to be the standard basis, so that $Av_i = A_{\cdot i}$ (the i -th column of A) and likewise for B . Then

$$\begin{aligned} (A \otimes B)(v_i \otimes v_j) &= (Av_i) \otimes (Bv_j) \stackrel{\text{std. basis}}{=} A_{\cdot i} \otimes B_{\cdot j} \\ &= \begin{pmatrix} A_{1i} B_{\cdot j} \\ \vdots \\ A_{mi} B_{\cdot j} \end{pmatrix} = \begin{pmatrix} \begin{pmatrix} A_{1i} B_{1j} \\ \vdots \\ A_{1i} B_{nj} \end{pmatrix} \\ \vdots \\ \begin{pmatrix} A_{mi} B_{1j} \\ \vdots \\ A_{mi} B_{nj} \end{pmatrix} \end{pmatrix}. \end{aligned} \quad (10)$$

Setting $v_i \otimes v_j \equiv w_k$ with $k = j + n i$ (for zero-based i), the matrix of $A \otimes B$ is the block matrix

$$\begin{aligned} A \otimes B &= \begin{pmatrix} A_{11} B_{\cdot 1} & \cdots & A_{11} B_{\cdot n} & \cdots & A_{1m} B_{\cdot 1} & \cdots & A_{1m} B_{\cdot n} \\ \vdots & & & & & & \vdots \\ A_{m1} B_{\cdot 1} & \cdots & A_{m1} B_{\cdot n} & \cdots & A_{mm} B_{\cdot 1} & \cdots & A_{mm} B_{\cdot n} \end{pmatrix} \\ &= \begin{pmatrix} A_{11} B & \cdots & A_{1m} B \\ \vdots & \ddots & \vdots \\ A_{m1} B & \cdots & A_{mm} B \end{pmatrix}, \end{aligned} \quad (11)$$

which is the Kronecker product for matrices.

1.4 References

- *Tensorprodukt* and *Kronecker-Produkt*, German Wikipedia.